



National University of Technology (NUTECH)

Course Plan – MATH 3901-Discrete Structures

CREDIT HOURS:

3 (Theory)

CONTACT HOURS:

48 (Theory)

Department:

Computer Engineering

SUBJECT DESCRIPTION:

The purpose of this course is to understand and use (abstract) discrete structures that are backbones of computer science. In particular, this class is meant to introduce logic, proofs, sets, relations, functions, counting, and probability, with an emphasis on applications in computer science.

SUBJECT OBJECTIVES:

Upon successfully completing this course, students will be able to:

1. Express and Analyze statements using formal logic to test their validity
2. Apply the basic properties and operations related to sets, relations and functions
3. To apply the concepts of number theory in encryption
4. Identify and use concepts of Combinatorics and graphs in different problems.

SUBJECT LEARNING OUTCOMES:

At the end of the course, students would be able to:

S#	CLO	Learning Domain	Taxonomy Level	PLO
1	<i>Interpret the key concepts of Discrete Structures such as Propositions, Sets, Relations, Functions, Algorithm growth etc.</i>	Cognitive	3	1
2	<i>Apply the concept of Number theory, Cryptography, Mathematical Induction and Combinations to solve problems</i>	Cognitive	3	1
3	<i>Use the idea of Relation, Closure, Equivalences and Graphs to handle problems of Discrete Structures</i>	Cognitive	3	1

TEXT AND MATERIAL:

Textbook (s)

1. Discrete Mathematics and Its Applications, 7th edition by Kenneth H. Rosen

Reference Books

2. Discrete Mathematics with Applications, 4th Edition by Susanna S. Epp
3. Discrete Mathematics, 7th edition by Richard Johnson Baugh
4. Discrete Mathematical Structures, 4th edition by Kolman, Busby & Ross
5. Discrete and Combinatorial Mathematics: An Applied Introduction by Ralph P. Grimaldi

	National University of Technology (NUTECH)
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RELEVANT PROGRAM LEARNING OUTCOMES (PLOs):

The course is designed so that students will achieve the following PLOs:

1	Academic Education	☒	6	Individual and team work	☐
2	Knowledge for solving computing Problems	☐	7	Communication	☐
3	Problem analysis	☐	8	Computing Professionalism and Society	☐
4	Design/Development of Solutions	☐	9	Ethics	☐
5	Modern Tool Usage:	☐	10	Lifelong Learning	☐

RELEVANCE OF COURSE TO REAL LIFE (PRACTICAL APPLICATIONS):

It's often said that mathematics is useful in solving a very wide variety of practical problems. focus on discrete mathematics, which, broadly conceived, underpins about half of pure mathematics and of operations research as well as all of computer science. As time goes on, more and more mathematics that is done, both in academia and in industry, is discrete. But what are the actual applications people talk about when they say discrete mathematics can be applied? What problems are being solved? A few are listed below

Computers run software and store files. The software and files are both stored as huge strings of 1s and 0s. Binary math is discrete mathematics.

Google Maps uses discrete mathematics to determine fastest driving routes and times. There is a simpler version that works with small maps and technicalities involved in adapting to large maps. Encryption and decryption are part of cryptography, which is part of discrete mathematics. For example, secure internet shopping uses public-key cryptography.

Designing password criteria is a counting problem: Is the space of passwords chosen large enough that a hacker can't break into accounts just by trying all the possibilities? How long do passwords need to resist such attacks? (find out here!)

ASSESSMENT SYSTEM:

There will be 04 Homework Assignments, 04 Quizzes, 01 PBL Activities, 01 Midterm and 01 Final Exam. Date of submission of assignments will be reflected. Late submission will have a penalty.

To encourage reading (reading assignments are reflected in course schedule) and discourage copying of homework assignments, all quizzes will be 50% from reading assignments and 50%

	National University of Technology (NUTECH)
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from problem sets in assignments. Relative grading system will be followed to award grades.

Weightages are as under:

<u>Theoretical/Instruction</u>	<u>No</u>	<u>Percentage</u>
Assignment	4	5%
PBLs	1	5%
Quiz	4	10%
Mid Semester Exam	1	30%
End Semester Exam	1	50%
Total :		100%

NATURE OF ASSESSMENTS

Time barred MCQs, Short questions/True-False/Numerical will be conducted to safeguard against cheating and to find copying and plagiarism at the time of marking. Exams will be conducted on campus. A minimum of three quizzes and assignments each (including a PBL) will be conducted.

CONDUCT AND SUBMISSION OF ASSESSMENTS

Students will be required to submit a hard copy of handwritten solution on A₄ paper within submission time. Students can also email the solution to the respective teacher.

OFFICE HOURS

Tuesday (10:35-03:45)

INSTRUCTOR:

Name: Dr. Muhammad Sajjad
Office: NUSASH, Academic Block, 4th floor (NUTL Office)
Email: muhammad.sajjad@nutech.edu.pk

INSTRUCTOR'S EXPERIENCE

Dr. Muhammad Sajjad is a dedicated mathematician from Pakistan, currently serving as an Assistant Professor at the National University of Technology (NUTECH), Islamabad. He earned his PhD from Quaid-i-Azam University Islamabad, Pakistan in 2024 and is an esteemed member of the American Mathematical Society (AMS). Recognized for his outstanding contributions to cybersecurity and cryptography, Dr. Sajjad has received several prestigious accolades, including two international research awards in cybersecurity and cryptography, the Computer Scientist Award from China, and the National Cryptology Award from NCCS Pakistan. His research spans multiple disciplines, including Mathematics, Electrical Engineering, and Computer Science, with a strong emphasis on Vector Algebras, Non-commutative and Non-Associative Algebras, Number Theory, Coding Theory, Channel



National University of Technology (NUTECH)

Course Plan – MATH 3901-Discrete Structures

Coding, Cryptography, Cryptology, Post-Quantum Cryptography, Graph Theory, and Elliptic Curves. His work is particularly focused on designing efficient error-correcting codes, advancing data security, and exploring complex algebraic structures. Dr. Sajjad is an active contributor to the global academic community, having delivered talks at prestigious international conferences. His research collaborations extend to leading universities in South America, Denmark, and Saudi Arabia, further strengthening his impact on the field. He undergone professional training in various aspects of academic management and possesses strong programming skills in Python, MATLAB, C++, and more. Over his academic career, He has taught 21 Bachelor and Master courses and published 13 research papers. Through his dedication to teaching, research, and innovation, Dr. Muhammad Sajjad continues to shape the future of mathematical sciences and information security.

DETAILED CONTENTS AND THEIR CONTRIBUTION TO PLOs:

Week	Topics Covered	Assessments	CLO No.	PBL	PLO No.
1	Logic and truth table basics	Quiz 1	1	CLO's (1+2+3)	1
2	Extend the understanding logic applications and Arguments		1		
3	Sets, Venn Diagrams, Set Identities	Asg. 1	1		
4	Relation, Functions, Types of functions, Inverses		1		
5	Number Theory	Asg. 2	1		
6	Mathematical Induction, MI for divisibility	Quiz 2	2		
7	Cryptography		2		
8	MSE		1+2		
9	Counting Principals	Asg. 3	2		
10	Basics, Pigeonhole principle, Combinations		2		
11	Relations, Properties of relations	Quiz 3	2		
12	Closure, Equivalence Relation		3		
13	Partially Order Set		3		
14	Graphs, Representing Graphs and Isomorphism's	Asg. 4	3		
15	Euler and Hamiltonian Graphs		3		
16	Graph Theory Applications in Cryptography	Quiz 4	3		

Prepared By:

Dr. Muhammad Sajjad (Instructor)

Approved By:

HOD CEN